Introduction to Natural Language Processing

HW 2

LLM-as-a-Judge: Predicting Human Preferences

Task introduction

- Building your own **LLM Judge** that evaluates and compares AI-generated responses. Your goal is to maximize agreement with ground-truth human preferences.
- You are given a dataset of pairwise conversations. Your task is to predict the **verdict** — which response is better:

Verdict	Meaning
A	Response 1 (dialog 1) is better
B	Response 2 (dialog 2) is better
tie	Both responses are equally good
neither	Both responses are equally bad

- Requirement :
  - Submit a report and your source code to E3
  - Upload your submission to Kaggle

Task rules & constraints

- **Model size limit:** Your model must have **≤ 9B** total parameters (including MoE models — total params, not active params). The model must be **open-weight** (e.g., Gemma-4-E4B-IT, Qwen3.5-9B, LLaMA-3.1-8B, Mistral-7B).
- **Any approach is allowed:** prompt engineering, fine-tuning, reinforcement learning (e.g., DPO/PPO), ensembles, post-processing calibration — anything goes, as long as you respect the **model size limit**.
- **API usage policy:** You may call inference LLM APIs (e.g., NVIDIA NeMo) only if the underlying model is **open-weight and within the 9B parameter** limit. You may **NOT** use LLM APIs backed by **closed-source models** (e.g. GPT-5, Claude, Gemini, etc.) for generating predictions. Using any API for data augmentation during training is allowed but must be disclosed in your report.

Dataset

The dataset consists of real human preference judgments collected from a platform where users compared two anonymous AI responses to the same instruction and voted for the better one.

- HW2_dataset

File	Data_nums	Description
train.json	4000	Labeled samples for training and experimentation
test.json	1000	Unlabeled samples for submission
sample_submission.csv	1000	Example submission file

Data Introduction

Each sample in train.json is a JSON object:

```
{
  "id": 1,
  "dialog_1": [
    {"role": "user", "content": "What is quantum computing?"},
    {"role": "assistant", "content": "Quantum computing is ..."}
  ],
  "dialog_2": [
    {"role": "user", "content": "What is quantum computing?"},
    {"role": "assistant", "content": "It's a type of computation ..."}
  ],
  "num_turns": 1,
  "verdict": "A"
}
```

- dialog_1 / dialog_2: Full conversation history as a list of message objects. Each message has a role ("user" or "assistant") and content. In multi-turn conversations (num_turns > 1), user and assistant messages alternate.
- num_turns: Number of dialogue turns.
- verdict: The ground-truth label. **Only present in train.json.**

- **Note:** test.json has the same format but without the verdict field.

Kaggle Submission (70%)

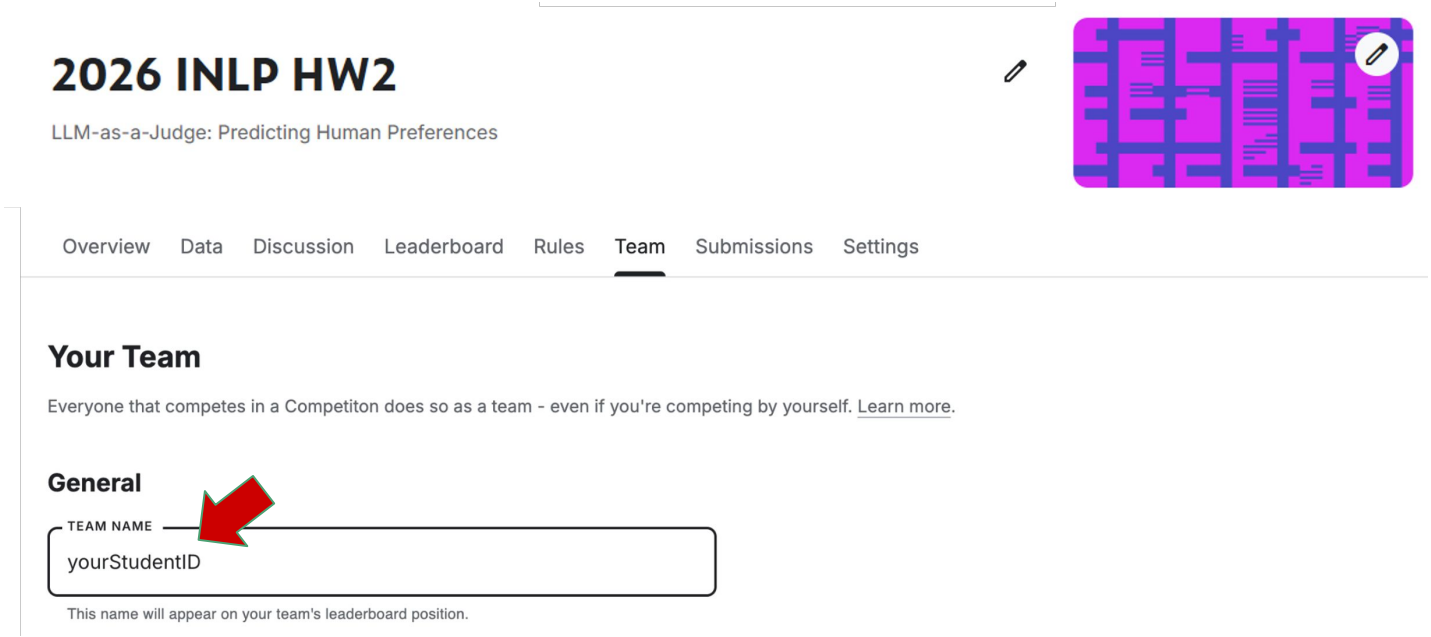
- Kaggle link
  Use your @nycu.edu.tw email address to join the competition
- Display team name : <student ID>
  - team name error : -5%
- Submission format
  - A 1001*2 .csv file, id starts from 1.
  - Prediction
    - A: Response 1 (dialog 1) is better
    - B: Response 2 (dialog 2) is better
    - tie: Both responses are equally good
    - neither: Both responses are equally bad
  - Column name must be id and verdict.
- There are one simple baseline and one strong baseline, beat them to get the higher score.

This row means:
Your prediction for “question id 0” is that “Response 1 is better”.

id	verdict
1	A
2	A
3	A
4	A
5	A
6	A
7	A

Change your team name

Please remember to change the team name to <student ID>, or there will be a deduction of 5 points for HW 2.



Kaggle Submission (70%)

- Public leaderboard is calculated with approximately 30% of the test data, private leaderboard is calculated with the other 70%, so the final standings may be different. You can only view your private leaderboard score after the competition has ended.
- Your submission is evaluated by Accuracy — the percentage of predictions that exactly match the ground-truth verdicts.
- You can submit at most 5 times each day and choose 3 of the submissions to be considered for the private leaderboard, or will otherwise default to the best public scoring submissions.

Report Submission

Please answer the following 3 questions, provide your thinking process as detailed as possible:

Q1. Method Description (10%)

- What model did you use and why?
- **Describe your approach and explain the reasoning in detail.**
- Show your exact prompt template(s).
- Detail any additional techniques (e.g., calibration, ensembling, fine-tuned, chain-of-thought).

Report Submission

Q2. Comparative Error Analysis (10%)

Implement at least 2 different judging approaches (e.g., zero-shot vs. chain-of-thought, different prompt designs, or different models) and compare their failure modes on the training set.

- Report each method's accuracy on a held-out split of train.json.
- For each method, identify its most common error pattern (e.g., errors concentrated on tie cases, long responses, ambiguous instructions) and provide one concrete example.
- Explain why different methods fail differently — what does each method's error pattern reveal about its underlying weakness?

Q3: Position Bias Analysis (10%)

Position bias is a well-known issue in pairwise LLM evaluation — models may systematically prefer the response in a particular position (first or second), regardless of quality.

- Design an experiment using the training data to measure whether your model exhibits position bias.
- Report quantitative results (e.g., how often does your model pick A vs B?).
- If you implemented any debiasing strategies, describe them and evaluate their effectiveness.

Please answer the questions in detail to get the full point of each question.

Grading policy

- Kaggle (70%)
  - 30% based on the public leaderboard score and 70% based on the private leaderboard score
  - Basic score :
    - Over strong baseline : 55
    - Over simple bassline : 40
    - Under simple baseline : 25
  - Ranking score:
    - 15-(15/N)*(ranking-1), N=numbers of people in the interval

- Report (30%)
  - 10% for each quesiton

You will receive 0 points if you do not submit the source code.

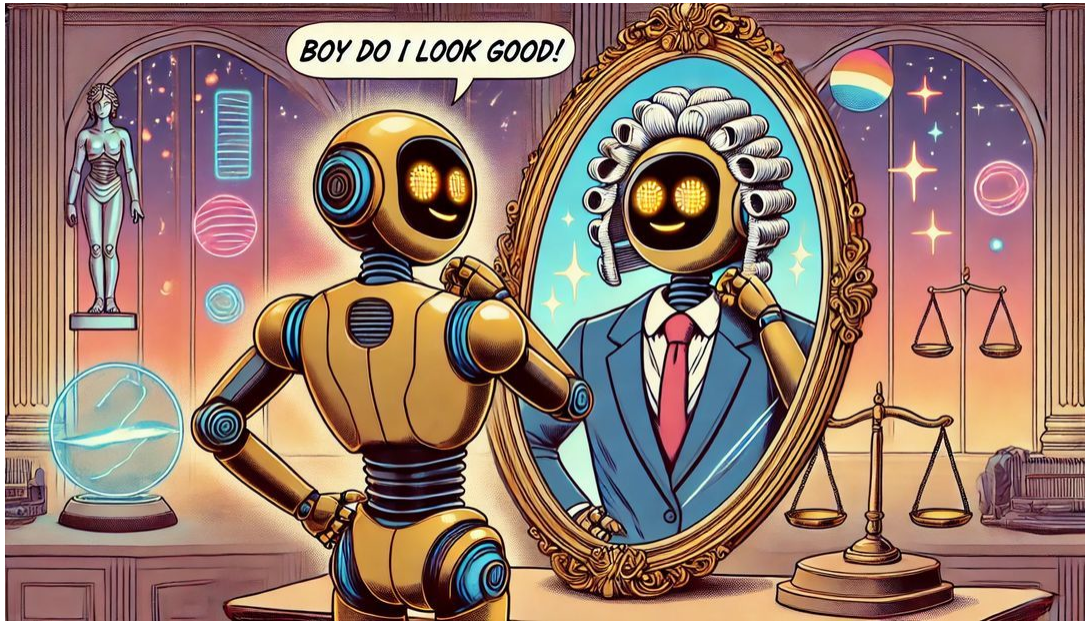
E3 Submission

Submit your source code and report to E3 before 5/5(Tue) 23:59, no late submissions will be accepted !

Format :

- HW2_<student ID>.zip
  - source code : HW2_<student ID>.py or HW2_<student ID>.ipynb
  - report : HW2_<student ID>.pdf

If you have any question about HW 2, please feel free to contact with TA: Wei-Lin Wen

through email weilin.cs14@nycu.edu.tw

Have Fun !